



US 20250280105A1

(19) **United States**
(12) **Patent Application Publication**
THODETI et al.
(10) **Pub. No.: US 2025/0280105 A1**
(43) **Pub. Date: Sep. 4, 2025**

(54) **METHODS AND APPARATUS FOR
MONITORING WIFI CAMERAS**

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(21) Appl. No.: **19/209,402**

(22) Filed: **May 15, 2025**

Related U.S. Application Data

(62) Division of application No. 18/743,200, filed on Jun. 14, 2024, which is a division of application No. 18/382,194, filed on Oct. 20, 2023, now abandoned,

which is a division of application No. 17/695,664, filed on Mar. 15, 2022, now abandoned.

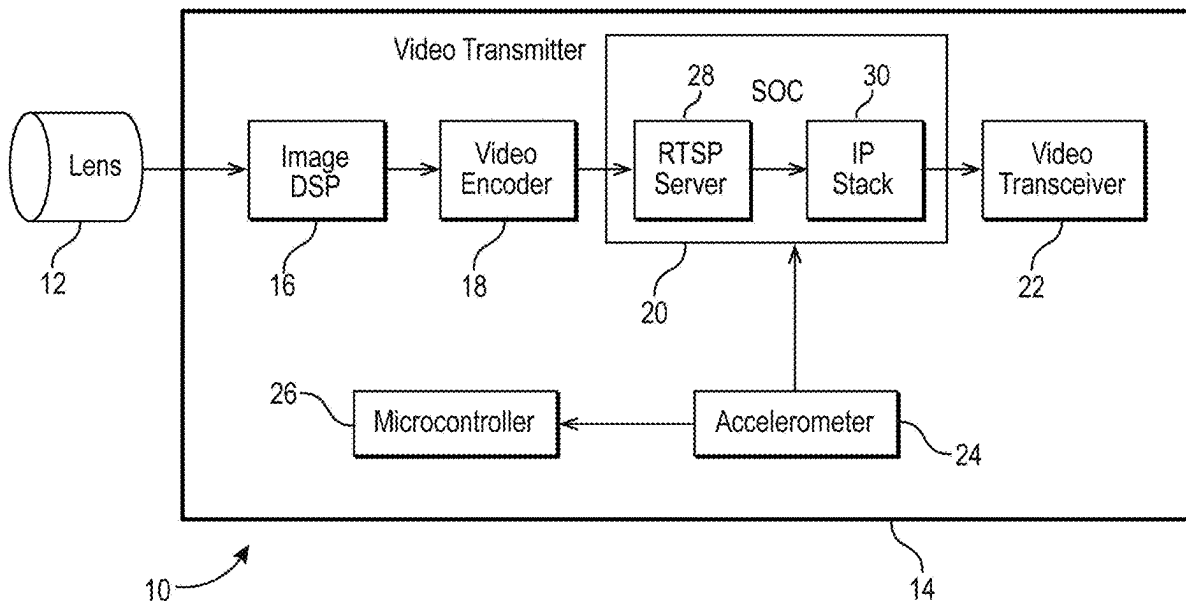
(60) Provisional application No. 63/193,988, filed on May 27, 2021.

Publication Classification

(51) **Int. Cl.**
H04N 17/00 (2006.01)
H04N 7/18 (2006.01)
H04N 23/52 (2023.01)
H04N 23/65 (2023.01)
(52) **U.S. Cl.**
CPC **H04N 17/002** (2013.01); **H04N 7/183**
(2013.01); **H04N 23/52** (2023.01); **H04N**
23/651 (2023.01)

(57) **ABSTRACT**

A method for detecting a freeze frame condition in a WiFi camera includes using the WiFi camera to capture a plurality of images. One of the images is compared to another one of the images. The freeze frame condition is detected as a result of determining that a level of similarity between the two captured images exceeds a threshold level.



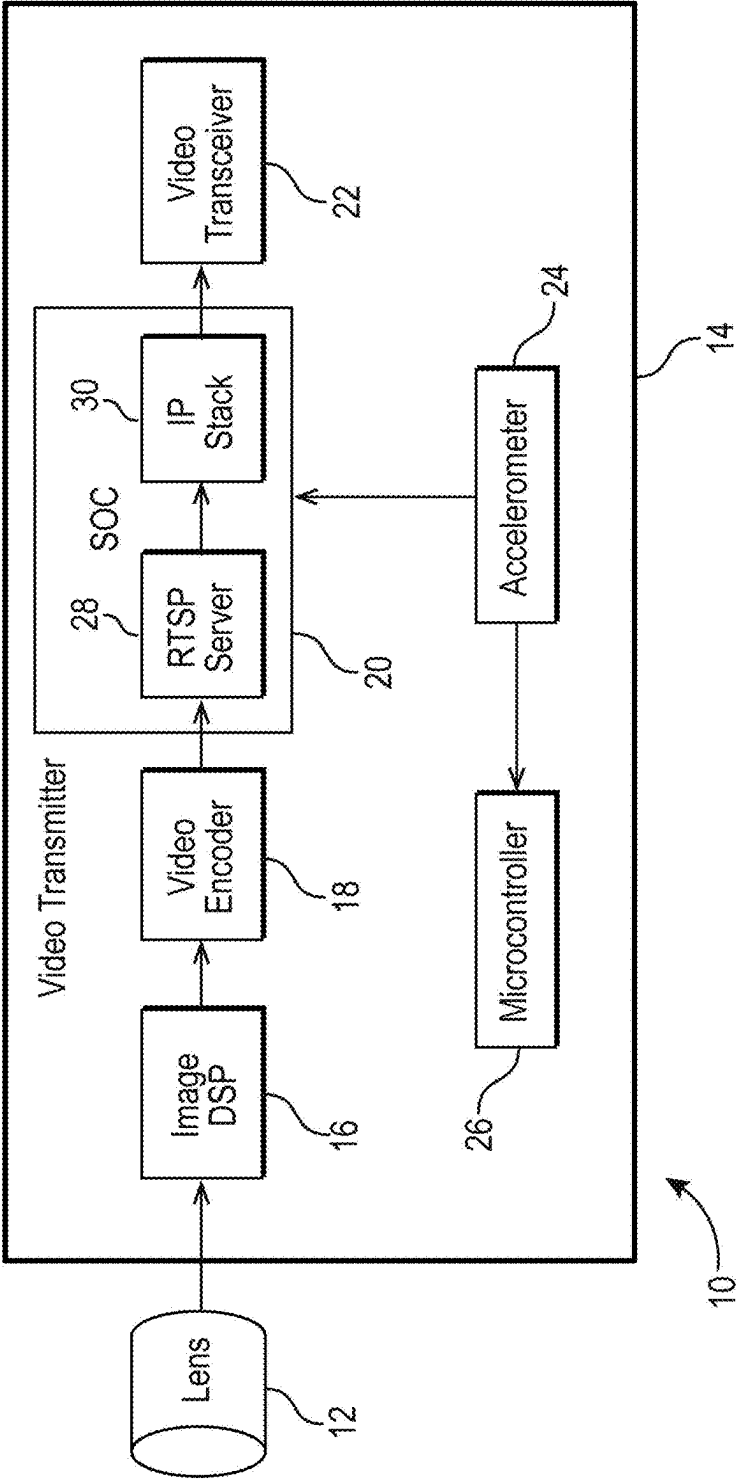


FIG. 1

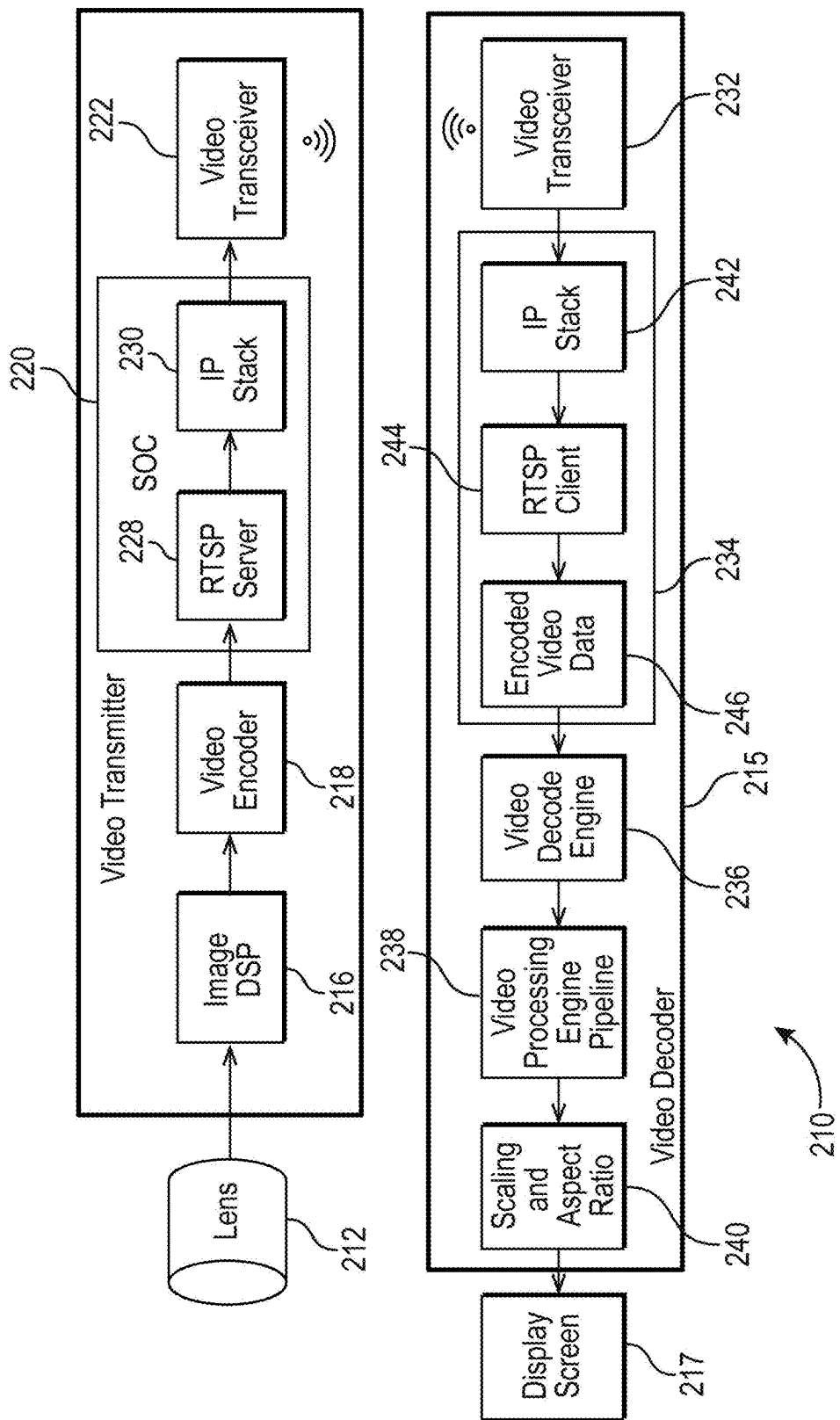


FIG. 2

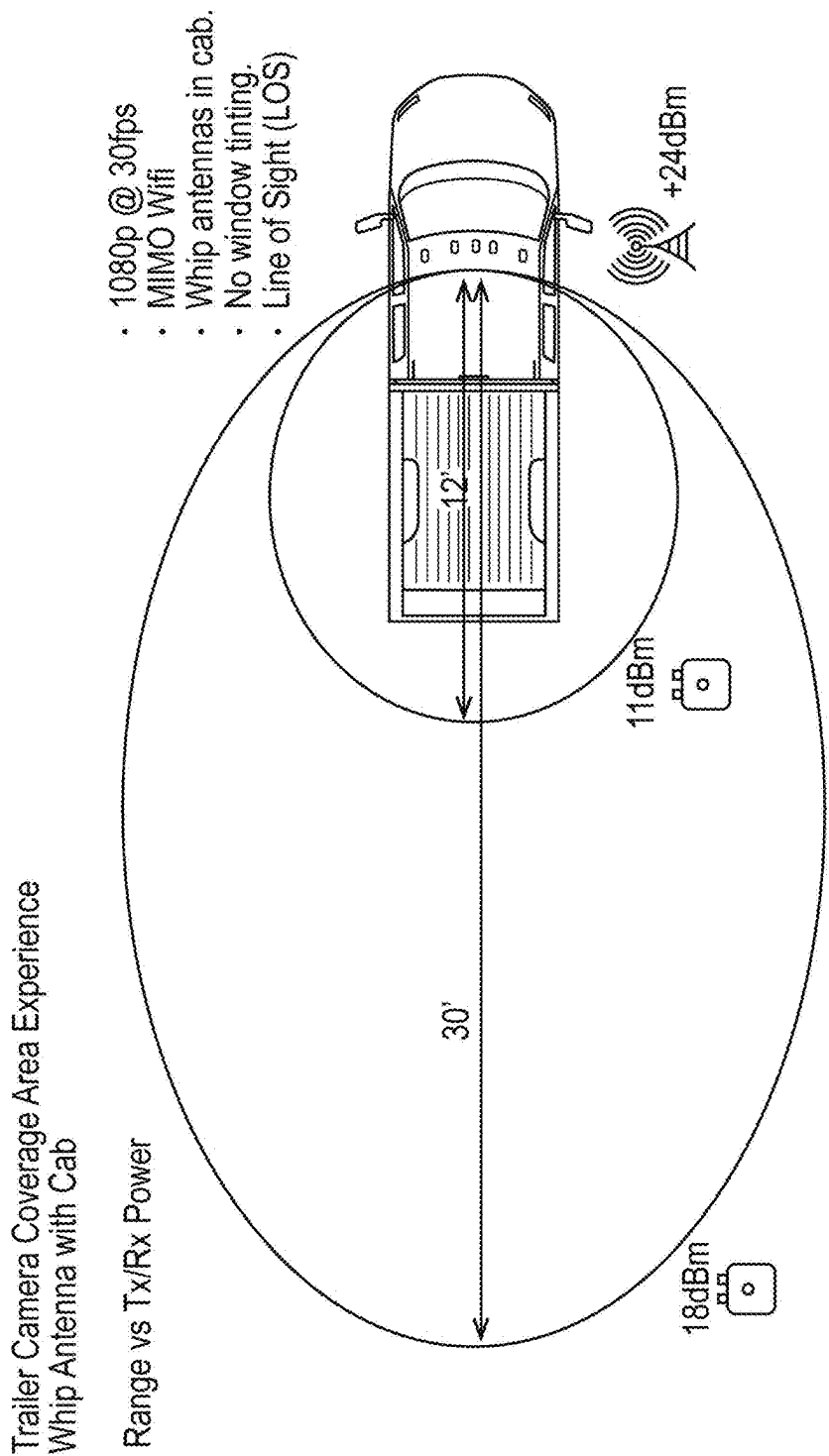


FIG. 3

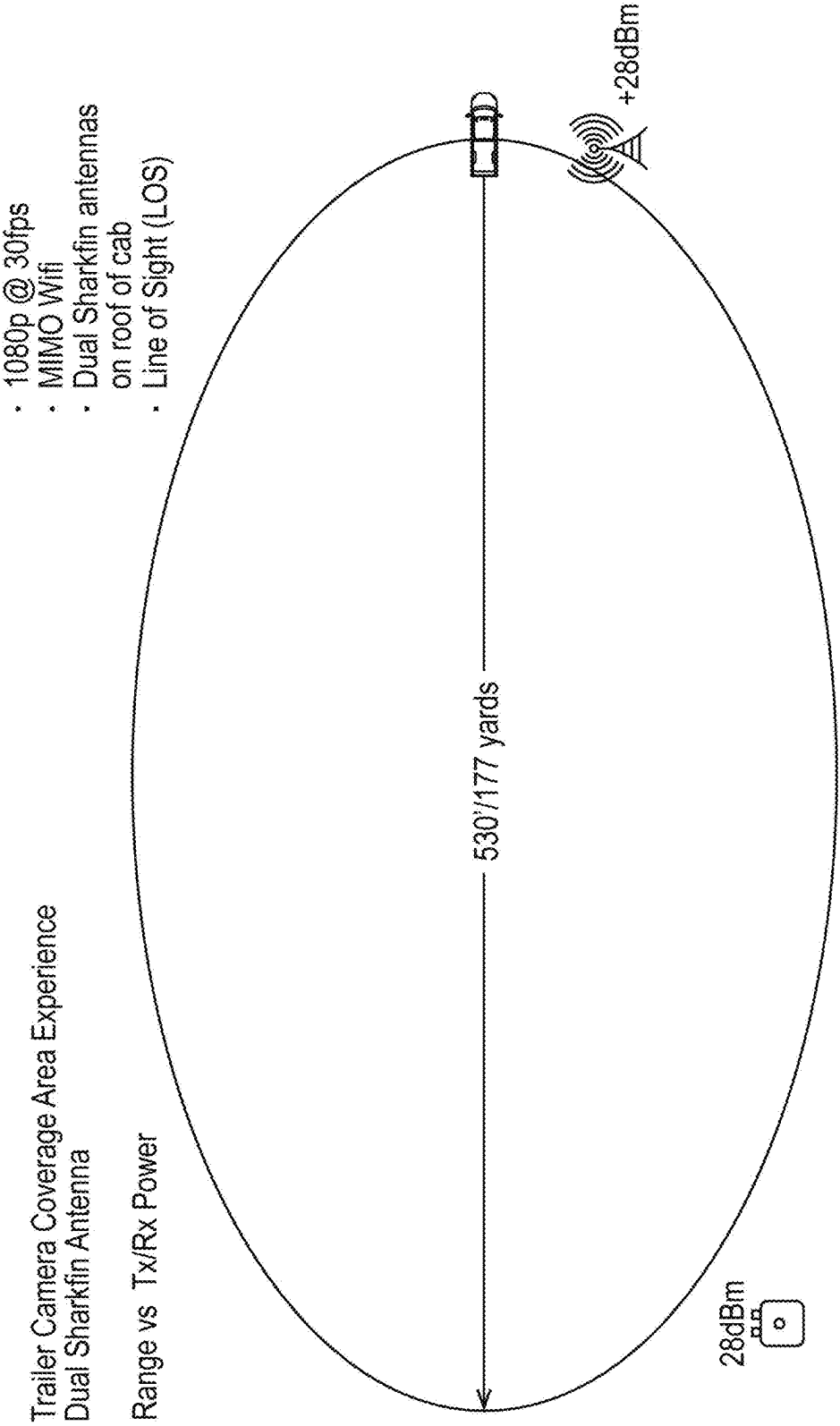
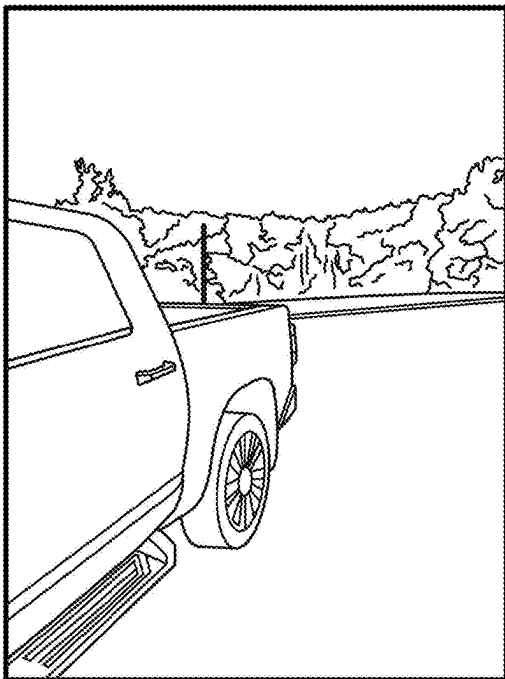


FIG. 4

Trailer Camera Coverage Area Experience
Dual Sharkfin Antenna

Picture Zoom 1x



Picture Zoom 10x

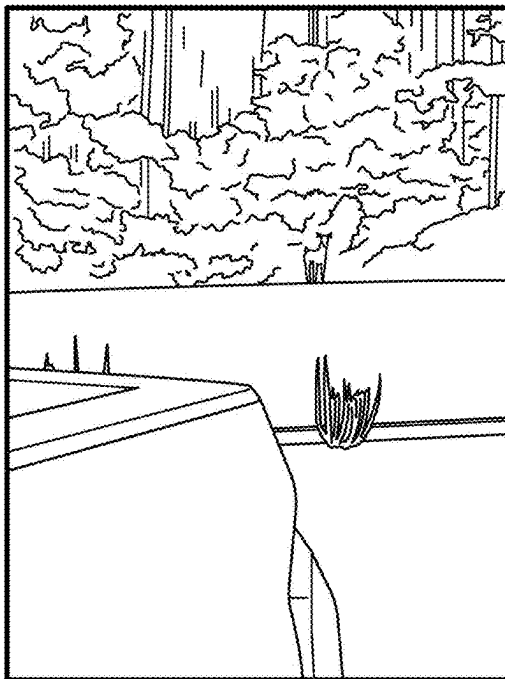


FIG. 5

Wi-Fi Camera Pairing Method

Wi-Fi camera scans directly the vehicles receiver info (Wi-Fi name and password) from vehicles screen.

User first activates Wi-Fi camera app on headunit, to get further directions for pairing.

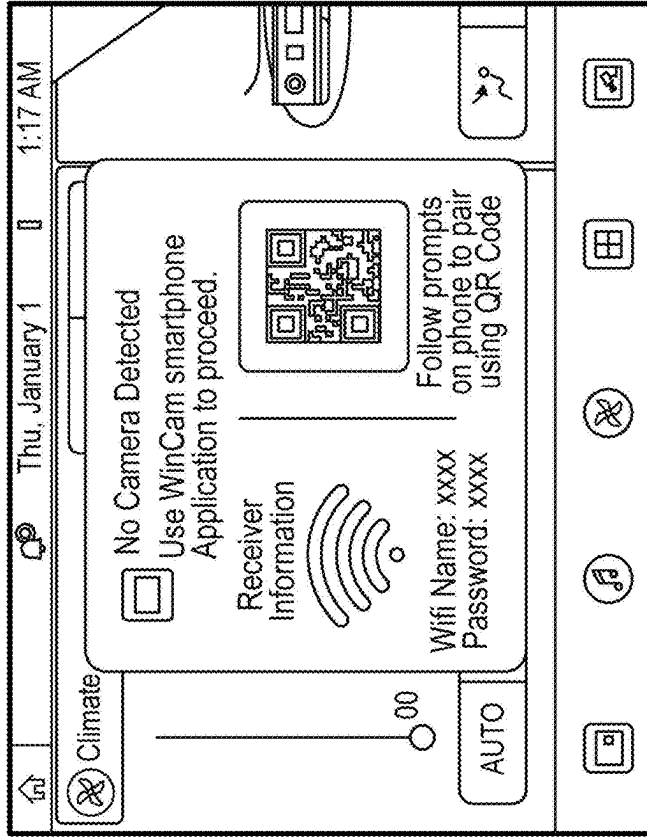
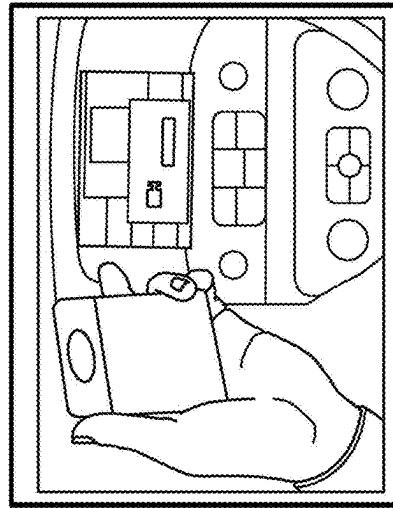


FIG. 6

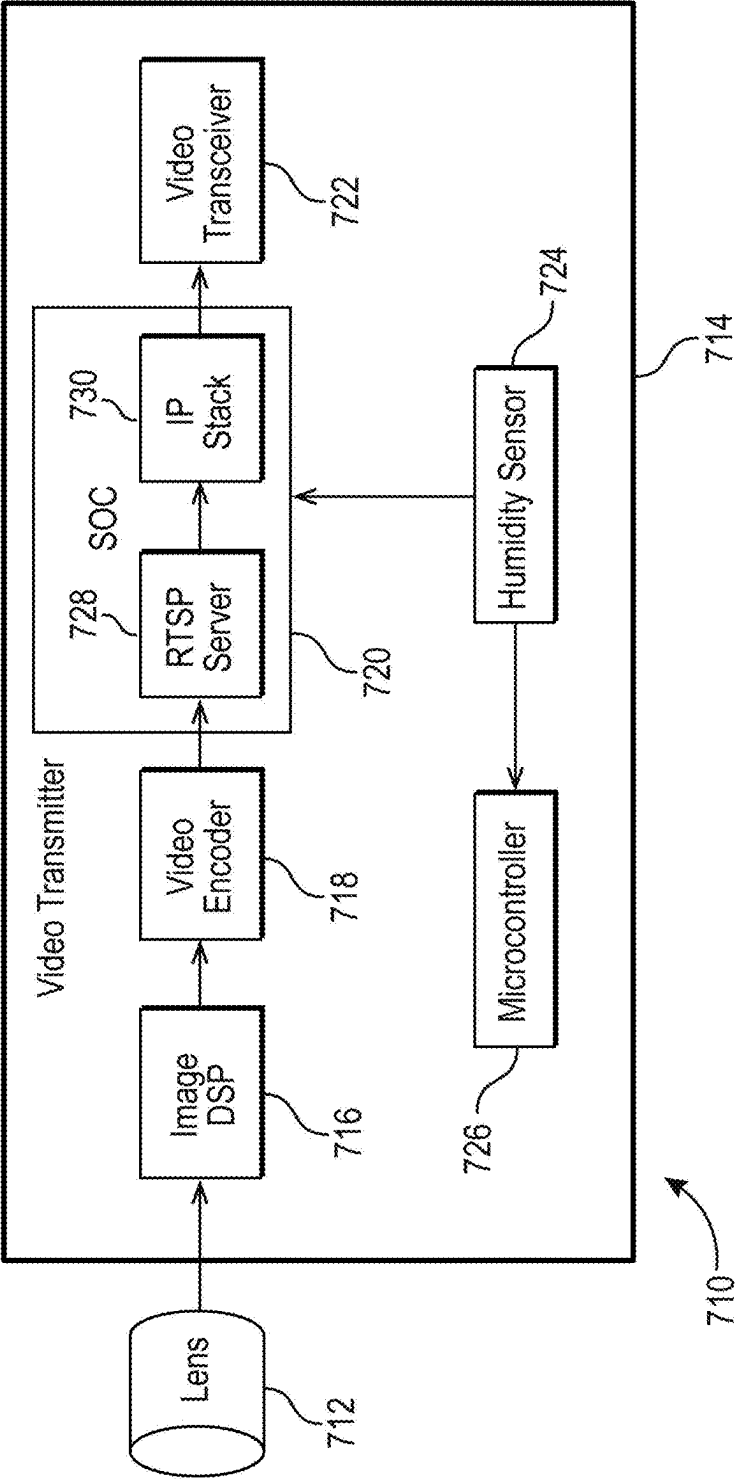


FIG. 7

WiFi Camera Modes of Operation

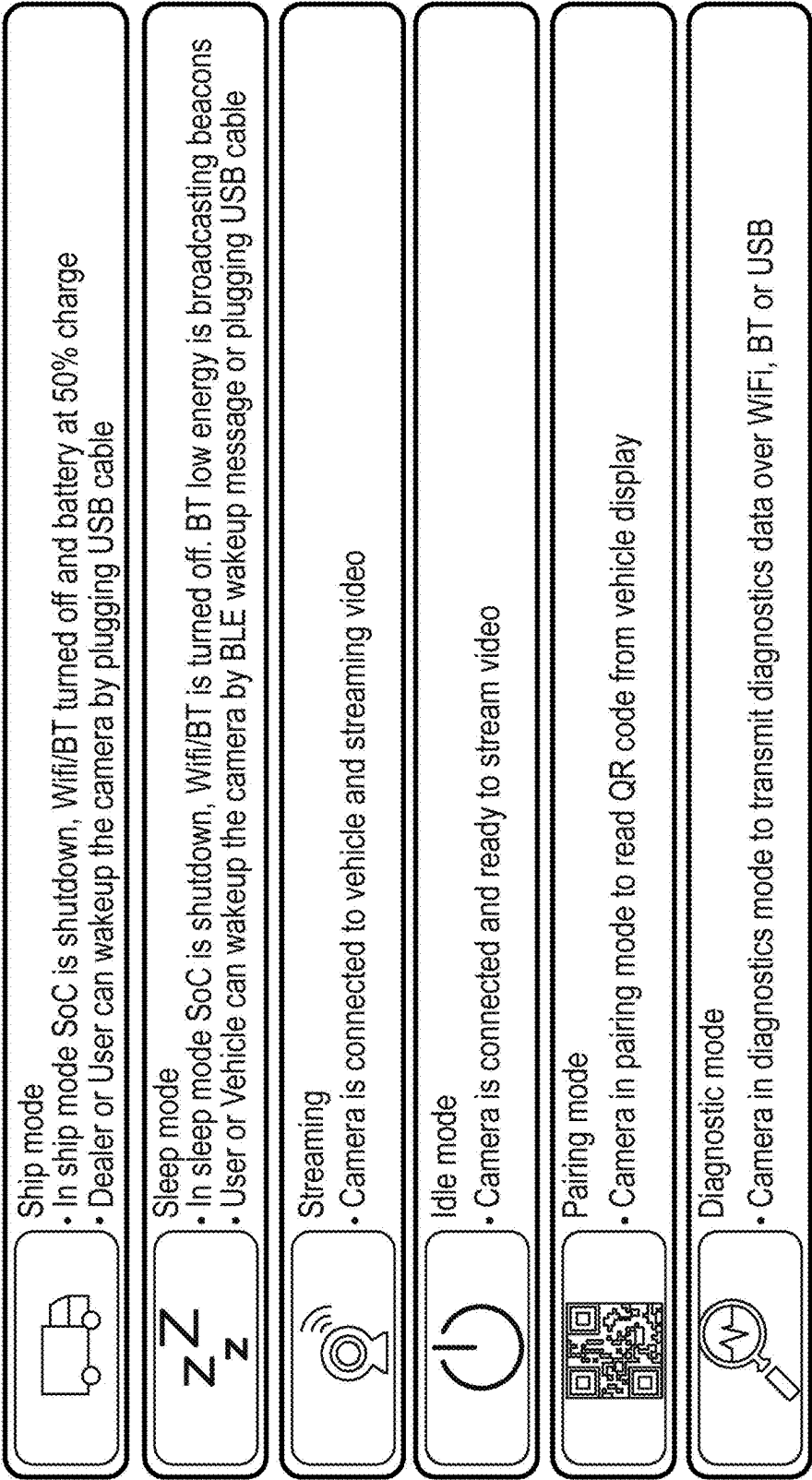
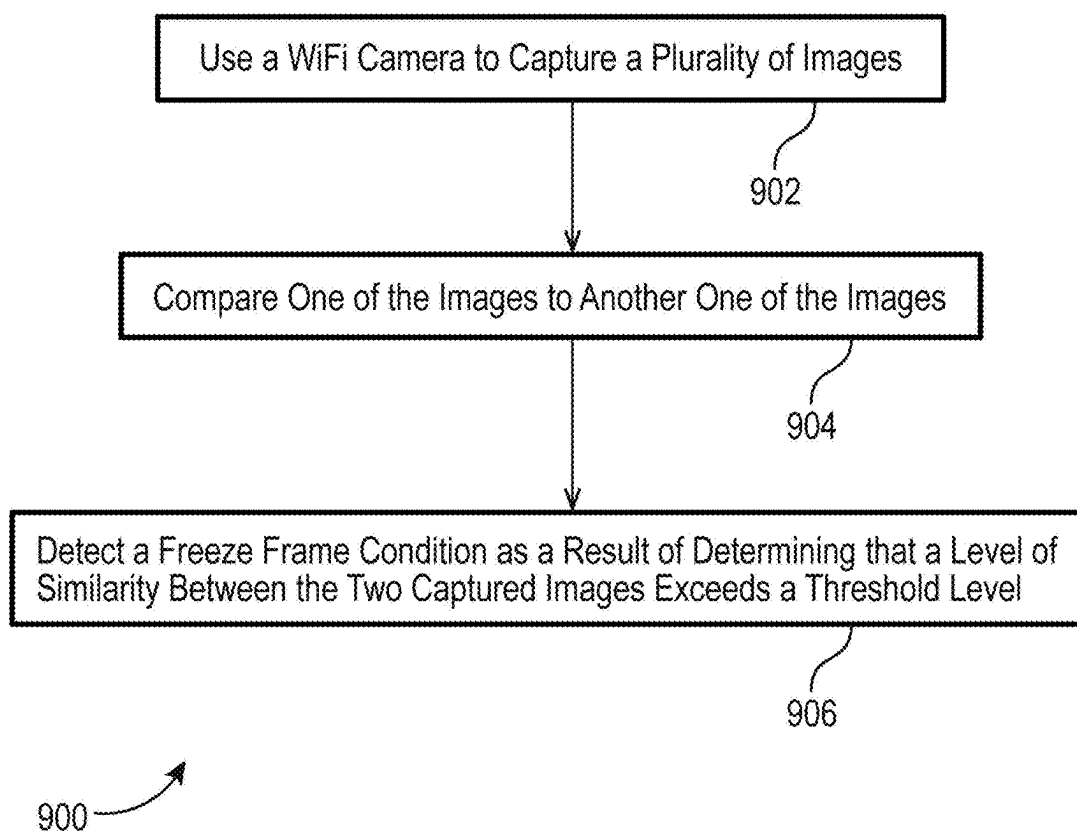


FIG. 8

**FIG. 9**

METHODS AND APPARATUS FOR MONITORING WIFI CAMERAS

CROSS-REFERENCED TO RELATED APPLICATIONS

[0001] This application is a divisional of U.S. patent application Ser. No. 18/743,200, filed on Jun. 14, 2024, which is a divisional application of U.S. patent application Ser. No. 18/382,194, filed on Oct. 20, 2023, which is divisional application of U.S. patent application Ser. No. 17/695,664, filed on Mar. 15, 2022, which claims benefit of U.S. Provisional Application No. 63/193,988, filed on May 27, 2021, the disclosures of which are hereby incorporated by reference in their entireties for all purposes.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to a shock monitoring system for a WiFi camera; freeze frame detection for a WiFi camera; video streaming wirelessly over long distances with low latency; WiFi camera pairing; humidity monitoring in a WiFi camera; and wireless IP camera low power operation.

2. Description of the Related Art

[0003] Wireless IP camera freeze frame detection with an in-vehicle infotainment system is unavailable. Wireless IP cameras for automotive and real-time video applications need a way to notify a user about stale video data or delayed video.

[0004] Another problem is that the available video stream bandwidth decreases over long distances. Low latency video streaming wirelessly over long distances is unavailable for automotive use-cases. Video streaming wirelessly for automotive use-cases is available but these does not support low latency and long distances necessary for trailer applications.

SUMMARY OF THE INVENTION

[0005] The present invention provides a method and apparatus for shock monitoring a WiFi camera for potential warranty issues. A log, timestamp and shock event may be recorded to mitigate potential warranty claims. With a record of shock events, any warranty claims may be voided based on the user mishandling the WiFi camera.

[0006] The present invention provides a method and apparatus for freeze frame detection on WiFi camera. A video buffer from a lens to digital signal processor and WiFi output is compared with a previous frame for 30 frames and triggers a notification to the user via video overlay or a message communication over WiFi/Bluetooth. Freeze frame detection can be implemented on in-vehicle infotainment system with less accuracy.

[0007] The present invention provides a method and apparatus for video streaming wirelessly over long distances with low latency for automotive and trailer applications. The invention may provide low latency video streaming wirelessly over long distances for automotive in vehicle infotainment display units. The inventive camera system automatically adjusts the video stream resolution and frame rate accordingly for a low latency video stream.

[0008] The present invention provides a method and apparatus for connecting a wireless internet protocol (IP) camera

with an in-vehicle infotainment system using a smartphone application and QR code display on the infotainment system.

[0009] The present invention provides a method and apparatus for humidity monitoring WiFi camera seal integrity. A log, timestamp and humidity leak event may be recorded, and the customer may be notified via a video signal receiver. Thus, damage to the WiFi camera internals may be mitigated.

[0010] The present invention provides a method and apparatus for wireless IP camera low power operation for sustaining on low energy for long duration. A wireless IP camera operating on battery power needs to conserve energy while not video streaming in order to sustain long duration before a battery recharge is required. While not video streaming, the wireless IP camera may shut down the main microprocessor, peripherals and high-power antennas to conserve energy and enable only Bluetooth low energy on a low power antenna for command and control.

[0011] It is possible to suspend video streaming and have the main microprocessor available for command and control. However, this method consumes more energy and requires the battery to be recharged more frequently.

[0012] The invention comprises, in one form thereof, a WiFi camera including an accelerometer and detecting mechanical shock events. An electronic processor is communicatively coupled to the accelerometer and produces a record of the shock events detected by the accelerometer.

[0013] The invention comprises, in another form thereof, a method for detecting a freeze frame condition in a WiFi camera. The WiFi camera is used to capture a plurality of images, compare one of the images to another one of the images, and detect the freeze frame condition as a result of determining that a level of similarity between the two captured images exceeds a threshold level.

[0014] The invention comprises, in yet another form thereof, a camera system automatically adjusting a video stream resolution and frame rate accordingly for a low latency video stream.

[0015] The invention comprises, in still another form thereof, a WiFi Camera Pairing Method including presenting a motor vehicle's receiver information on the vehicle's display screen. A Wi-Fi camera is used to directly scan the vehicle's receiver information on the vehicle's display screen.

[0016] The invention comprises, in a further form thereof, a WiFi camera including a humidity sensor detecting humidity levels. An electronic processor is communicatively coupled to the humidity sensor and produces a record of the humidity levels detected by the humidity sensor. A user is notified in response to the detected humidity levels exceeding a threshold.

[0017] The invention comprises, in another form thereof, a wireless IP camera that, in a non-video streaming mode, shuts down a main microprocessor, peripherals and high-power antennas to conserve energy and enable only Bluetooth low energy on a low power antenna for command and control.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The above-mentioned and other features and objects of this invention, and the manner of attaining them, will become more apparent and the invention itself will be better understood by reference to the following description

of embodiments of the invention taken in conjunction with the accompanying drawings, wherein:

[0019] FIG. 1 is a block diagram of one embodiment of a shock monitoring WiFi camera of the present invention.

[0020] FIG. 2 is a block diagram of one embodiment of a WiFi camera freeze frame detection arrangement of the present invention.

[0021] FIG. 3 is a schematic diagram of a Trailer Camera Coverage Area Experience Whip Antenna with Cab according to one embodiment of the invention.

[0022] FIG. 4 is a schematic diagram of a Trailer Camera Coverage Area Experience Dual Sharkfin Antenna according to one embodiment of the invention.

[0023] FIG. 5 is photographs illustrating a Trailer Camera Coverage Area Experience Dual Sharkfin Antenna according to one embodiment of the invention.

[0024] FIG. 6 is a schematic illustration of one embodiment of a WiFi Camera Pairing Method of the present invention.

[0025] FIG. 7 is a block diagram of one embodiment of a humidity monitoring seal integrity WiFi camera of the present invention.

[0026] FIG. 8 illustrates WiFi Camera Modes of Operation according to one embodiment of the present invention.

[0027] FIG. 9 is a flow chart of one embodiment of a method of the present invention for detecting a freeze frame condition in a WiFi camera.

DETAILED DESCRIPTION

[0028] The embodiments hereinafter disclosed are not intended to be exhaustive or limit the invention to the precise forms disclosed in the following description. Rather the embodiments are chosen and described so that others skilled in the art may utilize its teachings.

[0029] FIG. 1 illustrates one embodiment of a shock monitoring WiFi camera **10** of the present invention including a lens **12** and a video transmitter **14**. Video transmitter **14** includes an image digital signal processor **16**, a video encoder **18**, a systems-on-chip **20**, a video transceiver **22**, an accelerometer **24**, and a microcontroller **26**. Systems-on-chip **20** includes a real time streaming protocol server **28** and an internet protocol stack **30**.

[0030] During use, systems-on-chip **20** performs shock monitoring while camera **10** is in full operation, standby and idle modes. Microcontroller **26** performs shock monitoring while camera **10** is in low power and sleep modes.

[0031] FIG. 2 illustrates one embodiment of a WiFi camera freeze frame detection arrangement **210** of the present invention, including a lens **212**, a video transmitter **214**, a video decoder **215**, and a display screen **217**. Video transmitter **214** includes an image digital signal processor **216**, a video encoder **218**, a systems-on-chip **220**, and a video transceiver **222**. Systems-on-chip **220** includes a real time streaming protocol server **228** and an internet protocol stack **230**. Video decoder **215** includes a video transceiver **232**, a systems-on-chip **234**, a video decode engine **236**, a video processing engine pipeline **238**, and a scaling and aspect ratio block **240**. Systems-on-chip **234** includes an internet protocol stack **242**, a real time streaming protocol client **244**, and an encoded video data block **246**. Freeze frame detection by comparing with a previous video frame primarily involves lens **212**, image digital signal processor **216**, video encoder **218**, video decode engine **236**, video processing engine pipeline **238**, and display screen **217**.

[0032] FIG. 3 illustrates a Trailer Camera Coverage Area Experience Whip Antenna with Cab according to one embodiment of the invention.

[0033] FIG. 4 illustrates a Trailer Camera Coverage Area Experience Dual Sharkfin Antenna according to one embodiment of the invention.

[0034] FIG. 5 includes two photographs illustrating a Trailer Camera Coverage Area Experience Dual Sharkfin Antenna according to one embodiment of the invention.

[0035] FIG. 6 illustrates of one embodiment of a WiFi Camera Pairing Method of the present invention. A Wi-Fi camera scans directly the vehicle's receiver information (e.g., WiFi name and password) from the vehicle's display screen. The user first activates a Wi-Fi camera app on a head unit in order to get further directions for pairing.

[0036] FIG. 7 illustrates one embodiment of a humidity monitoring seal integrity WiFi camera **710** of the present invention including a lens **712** and a video transmitter **714**. Video transmitter **714** includes an image digital signal processor **716**, a video encoder **718**, a systems-on-chip **720**, a video transceiver **722**, a humidity sensor **724**, and a microcontroller **726**. Systems-on-chip **720** includes a real time streaming protocol server **728** and an internet protocol stack **730**.

[0037] During use, systems-on-chip **720** performs humidity monitoring while camera **10** is in full operation, standby and idle modes. Microcontroller **26** performs humidity monitoring while camera **10** is in low power and sleep modes.

[0038] FIG. 8 illustrates WiFi Camera Modes of Operation according to one embodiment of the present invention. In ship mode, the systems-on-chip is shut down, the WiFi/Bluetooth is turned off, and the battery is at 50% charge. The dealer or user can wake up the camera by plugging in a USB cable.

[0039] In sleep mode, the systems-on-chip is shut down, WiFi is turned off. Bluetooth broadcasts beacons at low energy. The user or vehicle can wake up the camera by Bluetooth low energy (BLE) wakeup message or plugging in a USB cable.

[0040] In streaming mode, the camera is connected to the vehicle and streams video.

[0041] In idle mode, the camera is connected and is ready to stream video.

[0042] In pairing mode, the camera reads a QR code from a vehicle display.

[0043] In diagnostics mode, the camera transmits diagnostics data over WiFi, Bluetooth or USB.

[0044] FIG. 9 illustrates one embodiment of a method **900** of the present invention for detecting a freeze frame condition in a WiFi camera. In a first step **902**, a WiFi camera is used to capture a plurality of images. For example, a camera including lens **212**, a video transmitter **214**, and a video decoder **215** may capture a plurality of images.

[0045] Next, in step **904**, one of the images is compared to another one of the images. For example, a video buffer from a lens to digital signal processor and WiFi output may be compared with a previous frame for 30 frames.

[0046] In a final step **906**, the freeze frame condition is detected as a result of determining that a level of similarity between the two captured images exceeds a threshold level. For example, if a pixel-to-pixel comparison of the two images shows that more than a threshold number or per-

centage of pixels are identical in the two images, then it may be determined that there is a freeze frame condition.

[0047] While this invention has been described as having an exemplary design, the present invention may be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

What is claimed is:

1. A method for detecting a freeze frame condition in a WiFi camera, the method comprising:

using the WiFi camera to capture a plurality of images;
comparing one of the images to another one of the images;
and

detecting the freeze frame condition as a result of determining that a level of similarity between the two captured images exceeds a threshold level.

2. The method of claim 1 wherein the camera includes a lens, an image digital signal processor, a video encoder, a video decode engine, and a video processing engine pipeline.

3. The method of claim 1 wherein the comparing step is performed by use of a display screen.

4. The method of claim 1 wherein the detecting step is performed by use of a display screen.

5. The method of claim 1 further comprising sending a notification of the freeze frame condition to a human user via a video overlay.

6. The method of claim 1 further comprising sending a notification of the freeze frame condition to a human user over WiFi/Bluetooth.

7. The method of claim 1 wherein the comparing step includes a pixel-to-pixel comparison of the two images.

8. The method of claim 7 wherein the detecting of the freeze frame condition is a result of determining that more than a threshold number or percentage of pixels are identical in the two images.

9. A WiFi camera freeze frame detection arrangement, comprising:

a WiFi camera configured to capture a plurality of images;
and

an electronic processor communicatively coupled to the WiFi camera and configured to:

compare one of the images to another one of the images; and

detect a freeze frame condition as a result of determining that a level of similarity between the two captured images exceeds a threshold level.

10. The arrangement of claim 9 wherein the WiFi camera includes a lens, an image digital signal processor, a video encoder, a video decode engine, and a video processing engine pipeline.

11. The arrangement of claim 9 further comprising a display screen communicatively coupled to the electronic processor, the comparing step including using the display screen.

12. The arrangement of claim 9 further comprising a display screen communicatively coupled to the electronic processor, the detecting step including using the display screen.

13. The arrangement of claim 9 wherein the electronic processor is configured to send a notification of the freeze frame condition to a human user via a video overlay.

14. The arrangement of claim 9 wherein the electronic processor is configured to send a notification of the freeze frame condition to a human user over WiFi/Bluetooth.

15. The arrangement of claim 9 wherein the electronic processor is configured to compare one of the images to another one of the images by performing a pixel-to-pixel comparison of the two images.

16. The arrangement of claim 15 wherein the electronic processor is configured to detect a freeze frame condition as a result of determining that more than a threshold number or percentage of pixels are identical in the two images.

17. A method for detecting a freeze frame condition in a camera, the method comprising:

using the camera to capture a plurality of images;
comparing one of the images to another one of the images
by performing a pixel-to-pixel comparison of the two images; and

detecting the freeze frame condition as a result of determining that more than a threshold number or percentage of pixels are identical in the two images.

18. The method of claim 17 wherein the camera includes a lens, an image digital signal processor, a video encoder, a video decode engine, and a video processing engine pipeline.

19. The method of claim 17 further comprising sending a notification of the freeze frame condition to a human user via a video overlay.

20. The method of claim 17 further comprising sending a notification of the freeze frame condition to a human user over WiFi/Bluetooth.

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